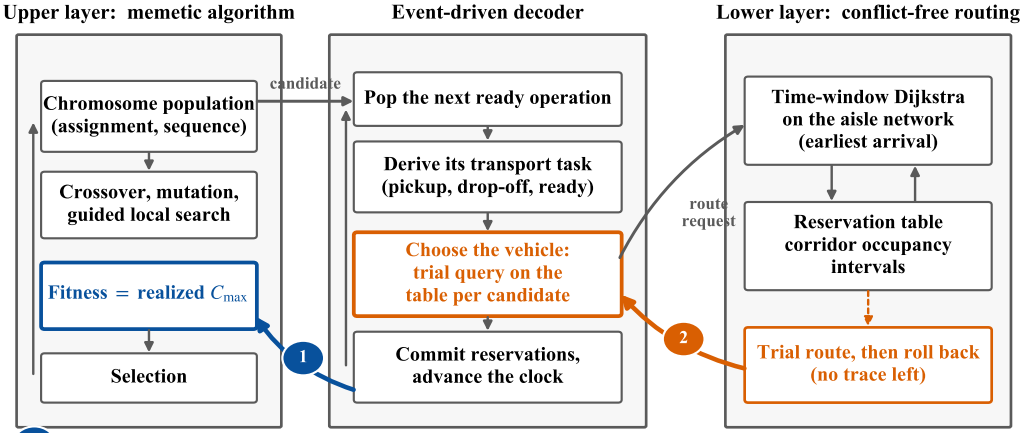
Upper layer: memetic algorithm
Event-driven decoder
Lower layer: conflict-free routing
Chromosome population (assignment, sequence)
Crossover, mutation, guided local search
Fitness = realized C_{max}
Selection
candidate
Pop the next ready operation
Derive its transport task (pickup, drop-off, ready)
Choose the vehicle: trial query on the table per candidate
Commit reservations, advance the clock
route request
Time-window Dijkstra on the aisle network (earliest arrival)
Reservation table corridor occupancy intervals
Trial route, then roll back (no trace left)
1
2
1 returns the conflict-free timetable as the fitness → once per candidate
2 returns the realizable arrival time → once per transport task